

b) $C \times B \times A$ $\{(0,1,a), (0,1,b), (0,1,c), (0,2,a), (0,2,b), (0,2,c), (1,1,a), (1,1,b), (1,1,c), (1,2,a), (1,2,b), (1,2,c)\}$

c) $C \times A \times B$ $\{(0,a,x), (0,a,y), (0,b,x), (0,b,y), (0,c,x), (0,c,y), (1,a,x), (1,a,y), (1,b,x), (1,b,y), (1,c,x), (1,c,y)\}$

d) $B \times B \times B$ $\{(x,x,x), (x,x,y), (x,y,x), (x,y,y), (y,x,x), (y,x,y), (y,y,x), (y,y,y)\}$

35) Find A^2 if a) $A = \{0, 1, 3\}$ b) $A = \{1, 2, a, b\}$
Solution: a) $A^2 = \{(0,0), (0,1), (0,3), (1,0), (1,1), (1,3), (3,0), (3,1), (3,3)\}$
 b) $A^2 = \{(1,1), (1,2), (1,a), (1,b), (2,1), (2,2), (2,a), (2,b), (a,1), (a,2), (a,a), (a,b), (b,1), (b,2), (b,a), (b,b)\}$

32) Suppose that $A \times B = \emptyset$, where A and B are sets. What can you conclude?
Solution: By Definition 8, $A \times B = \{(a,b) \mid a \in A \wedge b \in B\}$. Given, $A \times B = \emptyset$, i.e. by defn. of empty set, it has no elements. $\therefore (a,b) \notin \emptyset$, where $a \in A$ and $b \in B$. This is only possible when at least one of A or B must be a null set. $\square$

33) Let A be a set. Show that $\emptyset \times A = A \times \emptyset = \emptyset$
Solution: By Definition 8, $\emptyset \times A = \{(x,a) \mid x \in \emptyset \wedge a \in A\}$.
 By Defn. of null set, $\forall x (x \notin \emptyset)$. $\therefore \emptyset \times A$ must be an empty set, i.e. $\emptyset \times A = \emptyset$.

By Definition 8, $A \times \emptyset = \{(a,x) \mid a \in A \wedge x \in \emptyset\}$
 By Defn. of null set, $\forall x (x \notin \emptyset)$. $\therefore A \times \emptyset$ must be an empty set, i.e. $A \times \emptyset = \emptyset$.

40) Show that $A \times B \neq B \times A$, when A and B are nonempty, unless $A = B$.

Solution: Let, $A \neq B$. By Defn. 2, A and B do not contain the same elements. WLOG, let $x \in A$ but $x \notin B$. Since B is non-empty, it must contain at least one element. Let, $y \in B$. $\therefore$ By Defn. 8, $(x,y) \in A \times B$. By Defn. 8, $(p,q) \in (B \times A)$ if and only if $p \in B$ and $q \in A$. $\therefore x \notin B \therefore (x,y) \notin B \times A$. By Defn. 2, $A \times B \neq B \times A$. If $A = B$, then $A \times B = A \times A = B \times A$ $\square$